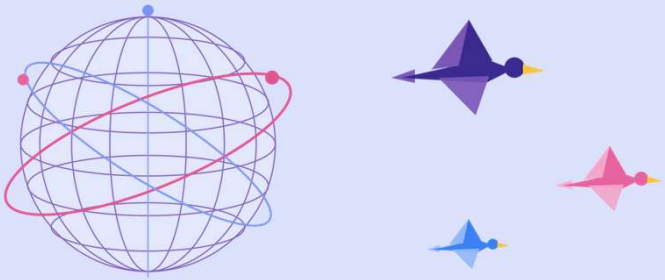
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details



Participant Details

PROJECT TITLE
Q-Rescue Next: An Adaptive and Fair Disaster Relief Allocation System Using Hybrid Quantum Computing and AI/ML

TRACK
4

TEAM NAME
BITNEXUS

TEAM LEADER
JAYALAKSHMI B

EMAIL / CONTACT
jayalakshmi8806@gmail.com

TEAM MEMBERS
POLIREDDY THRISHA REDDY – thrishareddy8@gmail.com

INSTITUTION / ORGANISATION
JAYALAKSHMI B – RIT
POLIREDDY THRISHA REDDY - DSU

COUNTRY
INDIA

Problem Statement

Problem:
During disasters, communities have different levels of damage, urgency, and resource requirements. Limited relief supplies such as food and medical kits must be allocated carefully, but traditional methods may not balance urgency, resource constraints, and fairness effectively.

Why it matters:
- Delayed relief can increase risks for vulnerable communities.
- Limited supplies may leave some communities with unmet needs.
- Data-driven allocation can help emergency teams make faster and more transparent decisions.

Proposed approach:
Use AI/ML-based risk assessment and hybrid quantum-classical optimization to support disaster relief resource allocation.

Objectives & Novelty

Objectives :

- Assess disaster risk using community-level data.
- Prioritize communities based on severity and medical urgency.
- Allocate limited food and medical kits within supply constraints.
- Compare classical and quantum-based optimization approaches.
- Visualize allocations, shortages, and fairness through a dashboard.

Novelty :

- Combines risk assessment with resource optimization.
- Considers multiple communities and resource types.
- Includes fairness and unmet-need analysis.
- Supports adaptive allocation when disaster conditions or supplies change.

Solution & Quantum Advantage

Solution workflow :

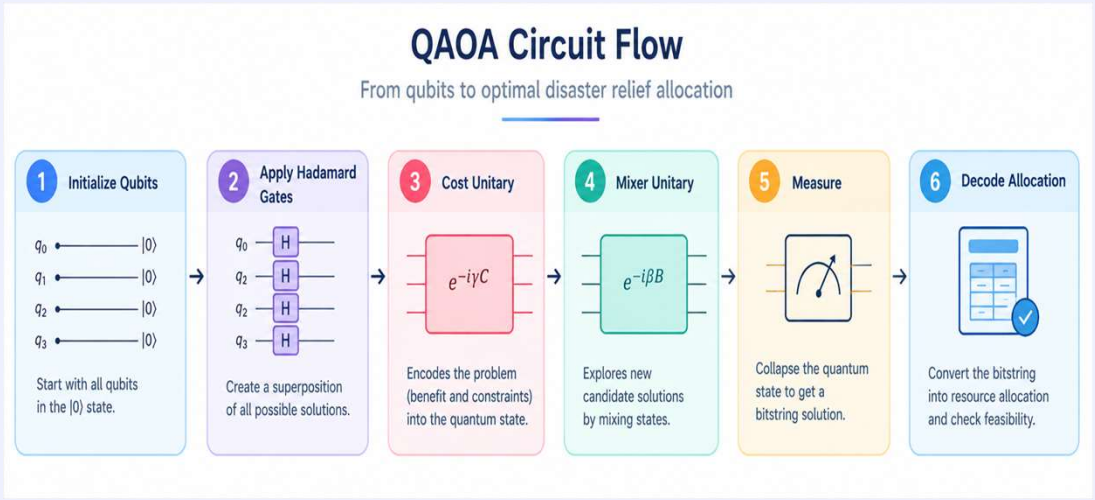
- Collect community and disaster data.
- Preprocess data and calculate risk scores.
- Formulate resource allocation as an optimization problem.
- Generate allocations using classical and QAOA-based methods.
- Validate constraints and evaluate fairness.
- Display results in the Streamlit dashboard.

Q-RESCUE NEXT

Quantum advantage :

- QAOA provides a quantum-classical approach for exploring candidate solutions to combinatorial optimization problems.
- Classical optimization provides a baseline for comparison.
- Performance will be evaluated using allocation quality, constraint satisfaction, runtime, and fairness.

Algorithm & Circuit Design



Input: Community risk scores, resource requirements, and supply limits.
Objective: Maximize priority-weighted relief while reducing unmet needs and respecting resource constraints.
Circuit: Parameterized quantum circuit with cost and mixer layers.
Optimization: A classical optimizer updates circuit parameters.
Measurement: Quantum circuit measurements generate candidate bitstrings.
Post-processing: Decode candidate allocations and check feasibility.

Implementation & Tech Stack

Component	Technology
Programming language	Python 3.11+
Dashboard	Streamlit
Risk assessment	Python ML model
Quantum framework	Qiskit
Quantum simulation	Qiskit Aer
Optimization	Classical optimizer and QAOA, if implemented
Data handling	CSV and Python
Testing	pytest
Development	VS Code
Version control	Git and GitHub

Implementation stages :

- Input validation and risk scoring
- Resource-allocation optimization
- Classical and quantum result evaluation
- Dashboard visualization
- CSV export and testing

Results & Analysis





6. Allocation Details

Community	Population	Risk Score	Food Needed	Food Allocated	Kits Needed	Kits Allocated	Severity	Medical Urgency
Riverdale	1200	244.0	600	498	70	59	5	5
Hillview	800	171.0	400	254	45	28	4	3
Greenfield	1500	175.0	750	607	60	44	3	4
Eastbank	1000	243.0	500	397	80	68	5	5
Meadowtown	600	98.0	300	44	30	1	2	2

Analysis

- The available food supply is insufficient to meet total community needs.
- Allocation results help identify shortages and communities requiring further support.
- Risk, severity, medical urgency, and fairness should be evaluated together.

Impact & Real-World Applications

Expected impact :

- Supports faster and more structured relief planning.
- Helps identify high-risk communities.
- Improves visibility of resource shortages.
- Encourages transparent, data-driven allocation decisions.

Applications :

- Flood and cyclone relief
- Earthquake emergency response
- Medical supply distribution
- Food and essential-kit allocation
- Disaster-response planning by government agencies and NGOs

Challenges & Future Scope

Challenges :

- Incomplete or inaccurate disaster data
- Limited relief supplies and competing community needs
- Balancing urgency, fairness, and allocation efficiency
- Quantum circuit size and simulator limitations
- Need for reliable comparisons between quantum and classical methods

Future scope :

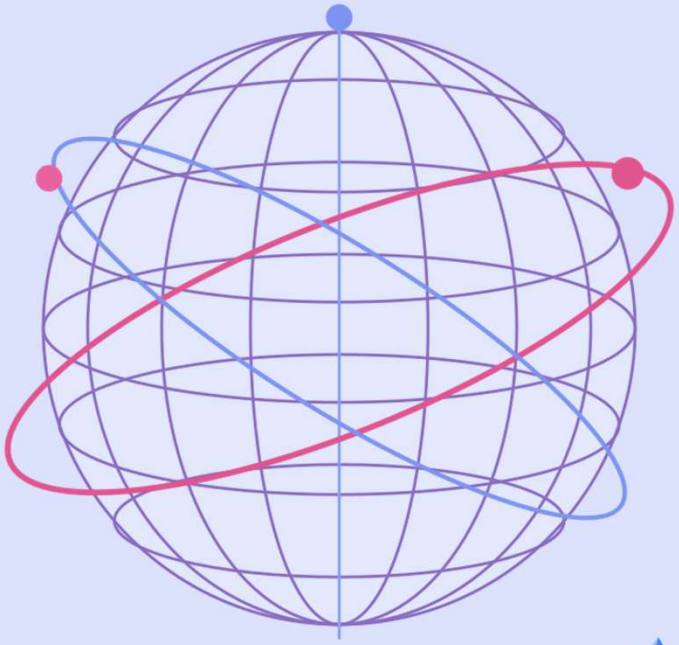
- Integrate real-time disaster and weather data.
- Improve risk prediction using trained ML models.
- Test QAOA against multiple classical optimization baselines.
- Add dynamic reallocation as conditions change.
- Evaluate performance on larger datasets and quantum hardware.
- Improve fairness metrics and explainability.

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



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